

Identification of Built-up and Water-Covered Regions Using Optical and SAR Observations

Independent optical and radar evidence mapped and cross-compared; one material conflict prevents a fused verdict

INVESTIGATION QUESTION

Use optical and sar images together to identify built-up and water-covered regions?

PRIMARY ASSESSMENT

Independent optical and SAR processing produced mapped built-up and water extents with area, region-count, and largest-contiguous-region metrics. However, a material cross-sensor conflict remains and prevents issuing a single fused classification; resolving this conflict requires a third observation. Users may employ the independent maps and the cross-sensor partition to guide decisions where single-sensor evidence is acceptable, but the unresolved area must be treated explicitly rather than incorporated into a fused product.

Investigation brief

The research objective, direct answer, and most decision-relevant facts are presented before methods or supporting imagery.

Objective

Assess whether the supplied optical and SAR observations can be combined to identify built-up and surface-water areas; evidence that would answer this is spatially resolved optical indices and radar backscatter classifications and a consistent late-fusion result showing corroboration across sensors.

Executive summary

The independent sensor analyses produced usable mapped findings. Optical Built-up ($NDBI > 0.05$) covers 1,933.2 hectares of the optical observation: 18.4% of the 10,510.2 hectares observed, in 5,143 regions. The largest contiguous region covers 401.1 hectares, with a mean Built-up of 0.14. Optical Water ($MNDWI > 0.15$) covers 4,812.2 hectares of the optical observation: 45.8% of the 10,510.2 hectares observed, in 197 regions. The largest contiguous region covers 4,502.7 hectares, with a mean Water of 0.69. Radar Built-up ($VV \geq -8$ dB and $VH \geq -15$ dB) covers 4,750.9 hectares of the radar observation: 45.2% of the 10,519.2 hectares observed, in 199 regions. The largest contiguous region covers 4,669.7 hectares, with a mean Built-up of -2.27. Radar Water ($VV \leq -17$ dB and $VH \leq -22$ dB) covers 1,694.2 hectares of the radar observation: 16.1% of the 10,519.2 hectares observed, in 2,028 regions. The largest contiguous region covers 523.3 hectares, with a mean Water of.

Question and direct answer

QUESTION	DIRECT ANSWER
Use optical and sar images together to identify built-up and water-covered regions?	The supplied observations produced separate, reviewable optical and radar findings; the final page records the boundary on a combined conclusion.

Key facts

Optical built-up area	Largest optical built-up region
Optical Built-up ($NDBI > 0.05$) covers 1,933.2 hectares of the optical observation: 18.4% of the 10,510.2 hectares observed, in 5,143 regions. Spectral-index analysis identified discrete built-up pixels across the optical scene totaling 1,933.2 hectares.	The largest contiguous region covers 401.1 hectares, with a mean Built-up of 0.14. One contiguous optical-built-up area dominates the optical-classified built-up extent and has a mean index value of 0.14.

Optical water area

Optical Water (MNDWI > 0.15) covers 4,812.2 hectares of the optical observation: 45.8% of the 10,510.2 hectares observed, in 197 regions.

The optical MNDWI threshold identified substantial surface-water extent in the optical scene.

Largest optical water region

The largest contiguous region covers 4,502.7 hectares, with a mean Water of 0.69.

Most optical-identified water area is contained in a single large contiguous region with mean MNDWI 0.69.

Complete validated findings

Every quantitative result remains verbatim from a validated claim. Interpretive text explains relevance without changing a value.

FINDING	VALIDATED RESULT AND INTERPRETATION
Optical built-up area	Optical Built-up (NDBI > 0.05) covers 1,933.2 hectares of the optical observation: 18.4% of the 10,510.2 hectares observed, in 5,143 regions. Spectral-index analysis identified discrete built-up pixels across the optical scene totaling 1,933.2 hectares.
Largest optical built-up region	The largest contiguous region covers 401.1 hectares, with a mean Built-up of 0.14. One contiguous optical-built-up area dominates the optical-classified built-up extent and has a mean index value of 0.14.
Optical water area	Optical Water (MNDWI > 0.15) covers 4,812.2 hectares of the optical observation: 45.8% of the 10,510.2 hectares observed, in 197 regions. The optical MNDWI threshold identified substantial surface-water extent in the optical scene.
Largest optical water region	The largest contiguous region covers 4,502.7 hectares, with a mean Water of 0.69. Most optical-identified water area is contained in a single large contiguous region with mean MNDWI 0.69.
Radar built-up area	Radar Built-up (VV >= -8 dB and VH >= -15 dB) covers 4,750.9 hectares of the radar observation: 45.2% of the 10,519.2 hectares observed, in 199 regions. Backscatter thresholds on calibrated SAR produced a larger radar-derived built-up extent than the optical-built-up result.
Largest radar built-up region	The largest contiguous region covers 4,669.7 hectares, with a mean Built-up of -2.27. Radar-derived built-up area is dominated by a single contiguous region whose mean backscatter-derived Built-up metric is -2.27.
Radar water area	Radar Water (VV <= -17 dB and VH <= -22 dB) covers 1,694.2 hectares of the radar observation: 16.1% of the 10,519.2 hectares observed, in 2,028 regions. SAR thresholds identified dispersed radar-quiet areas interpreted as water across many small regions.
Largest radar water region	The largest contiguous region covers 523.3 hectares, with a mean Water of -19.33. The largest radar-identified water body is 523.3 hectares with mean backscatter values at -19.33 dB.

Visual evidence

Each panel is interpreted against its linked validated finding; the image is evidence, not decoration.

Optical built-up classification



Built-up derived independently from optical evidence; unobserved ground is transparent.

WHAT THE IMAGE SHOWS

The optical analysis maps pixels meeting the $NDBI > 0.05$ threshold as built-up across the supplied optical observation.

WHAT IT SUPPORTS

It provides the spatial distribution and area metrics for optical-derived built-up used in the cross-sensor comparison.

EVIDENCE BASIS

The figure is rendered from retained spectral-index analysis output and is linked to a validated optical-built-up claim.

Radar built-up classification



Built-up derived independently from radar evidence; unobserved ground is transparent.

WHAT THE IMAGE SHOWS

The radar analysis maps pixels satisfying $VV \geq -8$ dB and $VH \geq -15$ dB as built-up across the supplied radar observation.

WHAT IT SUPPORTS

It provides the spatial distribution and area metrics for radar-derived built-up used in the cross-sensor comparison.

EVIDENCE BASIS

The figure is rendered from retained SAR backscatter processing output and is linked to a validated radar-built-up claim.

Visual evidence

Each panel is interpreted against its linked validated finding; the image is evidence, not decoration.

Optical water classification



Water derived independently from optical evidence; unobserved ground is transparent.

WHAT THE IMAGE SHOWS

The optical analysis maps pixels meeting the $MNDWI > 0.15$ threshold as surface water in the supplied optical observation.

WHAT IT SUPPORTS

It provides the spatial distribution and area metrics for optical-derived water used in the cross-sensor comparison.

EVIDENCE BASIS

The figure is rendered from retained spectral-index analysis output and is linked to a validated optical-water claim.

Radar water classification



Water derived independently from radar evidence; unobserved ground is transparent.

WHAT THE IMAGE SHOWS

The radar analysis maps pixels satisfying $VV \leq -17$ dB and $VH \leq -22$ dB as water in the supplied radar observation.

WHAT IT SUPPORTS

It provides the spatial distribution and area metrics for radar-derived water used in the cross-sensor comparison.

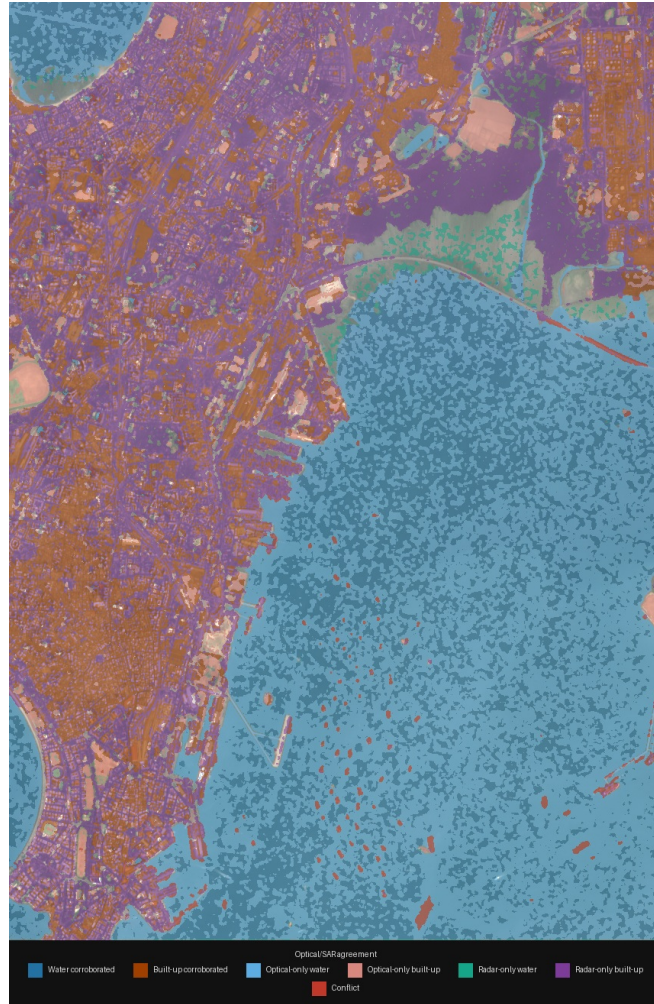
EVIDENCE BASIS

The figure is rendered from retained SAR backscatter processing output and is linked to a validated radar-water claim.

Visual evidence

Each panel is interpreted against its linked validated finding; the image is evidence, not decoration.

Cross-sensor agreement and conflict



Late-fusion partition for material regions of at least 5 ha; conflict requires a third observation.

WHAT THE IMAGE SHOWS

The synthesis map partitions the common study area into corroborated regions, single-sensor detections, and spatial conflicts for material regions of at least 5 ha.

WHAT IT SUPPORTS

It shows the spatial relationship between independent sensor interpretations and identifies where a fused conclusion is supported or prevented.

EVIDENCE BASIS

The figure is rendered from the retained late-fusion ledger and is linked to the validated claim that a material conflict requires a third observation.

Models and analytical roles

This method record explains what each analytical component examined, why it was selected, and how its output contributed to the result.

METHOD	ANALYTICAL ROLE	WHY USED
Image co-registration Version 1.3.0	The spatial grids of the supplied observations. Recorded the residual alignment used by downstream analysis.	To ensure the same map location was compared across inputs.
Spectral-index analysis Version 1.4.0	Multispectral reflectance relationships associated with the requested land-surface classes. Produced the optical classification layer and its measurements.	Spectral indices provide deterministic, interpretable evidence from optical bands.
SAR backscatter processing Version 1.3.0	Calibrated radar backscatter after speckle filtering and terrain correction. Produced the radar evidence used in the comparison.	Radar supplies an independent observation that is not based on visible colour.
Spatial late-fusion ledger Version 1.0.0	The location-by-location relationship between independently derived optical and radar evidence. Catalogued corroborated, single-sensor, conflicting, and non-informative regions.	Late fusion preserves agreement and conflict instead of concealing disagreement in a blended score.

How the evidence was assembled

Inputs were validated before specialist analysis. Each analytical component then produced a retained result with a model or method record. Claims were constructed from those outputs and linked to visible evidence. Only after validation did the language model organise the findings into this reader-facing explanation.

Interpretation boundary

Specialist and deterministic methods produced the retained evidence. The language model organised and explained those validated results; it did not measure pixels, alter geometry, or create quantitative findings.

Limitations and final assessment

The final page separates what the evidence established from what must remain unresolved.

Question and answer

QUESTION	DIRECT ANSWER
Use optical and sar images together to identify built-up and water-covered regions?	The supplied observations produced separate, reviewable optical and radar findings; the final page records the boundary on a combined conclusion.

What could not be fully grounded

AERIS cannot produce a fused conclusion: 1 material optical/SAR conflict requires a third observation. The individual sensor findings remain usable, but the unresolved relationship must not be presented as a single fused classification.

Final assessment

Independent optical and SAR processing produced mapped built-up and water extents with area, region-count, and largest-contiguous-region metrics. However, a material cross-sensor conflict remains and prevents issuing a single fused classification; resolving this conflict requires a third observation. Users may employ the independent maps and the cross-sensor partition to guide decisions where single-sensor evidence is acceptable, but the unresolved area must be treated explicitly rather than incorporated into a fused product.

Reproducibility note

The companion manifest retains input hashes, model versions, processing parameters, claim-to-evidence links, geometries, and render specifications. Transfer identifiers are intentionally excluded from this reader-facing document.